

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0640

George M Sullivan Generation Plant 2 100 kV
November 16, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0640. The Interconnection Customer proposes a 191.8 MW wind facility interconnecting at the George M Sullivan Generation Plant 2 100 kV substation in the Illinois area. The request was received on July 04, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,628,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pine Bluff South – George M Sullivan Generation Plant 2 100 kV	George M Sullivan Generation Plant 2 – Crum Canyon 100 kV	156.2	109	100.0
Crum Canyon – George M Sullivan Generation Plant 2 100 kV	George M Sullivan Generation Plant 2 – Pine Bluff S 100 kV	137.8	123	100.0
N Charlotte – Navy 1 100 kV	Army Depot Substation – Zion Energy Ctr 100 kV	116.0	136	10.8
Stapley – Rosemore B 100 kV L	Rosemore B – Stapley Substation 100 kV Circuit 1	108.6	159	6.3
Goose Haven – Golah 100 kV Circuit 2	Golah – Goose Haven 100 kV Circuit 1	107.0	182	5.9
Stapley – Rosemore B 100 kV Line 1	Rosemore B – Stapley 100 kV L	106.6	162	6.3
Golah Substation – Goose Haven 100 kV Circuit 1	Golah – Goose Haven 100 kV	104.1	187	5.9
Rosmore B – Goose Haven 100 kV	Goose Haven – Rosemore B 100 kV Line 1	100.8	204	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pine Bluff South – George M Sullivan Generation Plant 2 100 kV	Rebuild line with higher-capacity conductor	\$9,360,000	100% (\$9,360,000)
Crum Canyon – George M Sullivan Generation Plant 2 100 kV	Rebuild line with higher-capacity conductor	\$4,535,000	100% (\$4,535,000)
N Charlotte – Navy 1 100 kV	Rebuild line with higher-capacity conductor	\$1,194,000	55% (\$653,000)
Stapley – Rosemore B 100 kV Line 2	Reconductor line	\$1,000,000	100% (\$1,000,000)
Goose Haven – Golah 100 kV Circuit 2	Reconductor line	\$605,000	100% (\$605,000)
Stapley – Rosemore B 100 kV #1	Reconductor line	\$1,293,000	100% (\$1,293,000)
Goose Haven – Golah 100 kV Cir. 1	Reconductor line	\$588,000	100% (\$588,000)
Rosmore B – Goose Haven 100 kV	Reconductor line	\$594,000	100% (\$594,000)
	Total allocated to project		\$18,628,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.